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Noise-Based Wireless Communication Using Software-Defined Radio

bachelor's thesis

undergraduate academic programme
first cycle
computer science and mathematics

Supervisor: Assoc. Prof. Veljko Pejović, PhD
Co-supervisor: Assist. Prof. Areeb Ahmed, PhD

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Title: Noise-Based Wireless Communication Using Software-Defined Radio
Type of thesis: Bachelor's thesis, undergraduate academic programme in Computer Science and Mathematics
Supervisor: Assoc. Prof. Veljko Pejović, PhD
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Description:
This bachelor's thesis presents wireless communication based on transferring information through noise modulation. It explains the mathematical foundations of alpha-stable noise and develops a communication system based on such noise. The required modules are implemented in GNU Radio to enable an alpha-stable noise-based wireless link. Finally, the link is demonstrated and evaluated in a real-world environment.

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Description:
In this thesis, the candidate introduces wireless communication based on information transfer through noise modulation. The candidate presents the mathematical foundations of alpha-stable noise and devises a communication system based on such noise. The concept is implemented in GNU Radio. Finally, the candidate demonstrates and evaluates an alpha-stable noise-based link in a real-world environment.

Upon completing this bachelor's thesis, I would like to sincerely thank my supervisor, Assoc. Prof. Veljko Pejović, PhD, and my co-supervisor, Assist. Prof. Areeb Ahmed, PhD, for their expertise, guidance, help, and support throughout its preparation.

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Contents

Summary

Abstract

1 Introduction **1**

2 Wireless Communication and Its Challenges **4**

2.1 Fundamentals of Wireless Communication 4

2.2 Existing Wireless Communication Technologies 6

2.3 Key Challenges of Modern Wireless Communication Systems . 8

2.4 Noise-Based Communication 10

3 α-Stable Distributions **11**

3.1 Stable Distributions 11

3.2 Definition of α-Stable Distributions 12

3.3 Parameters of the α-Stable Distribution 15

3.4 Special Cases of α-Stable Distributions 17

3.5 Generalised Central Limit Theorem 20

3.6 Significance of Stability for Communication-Channel Modelling 23

4 Communication Using α-Stable Noise **25**

4.1 Encoding Information 25

4.2 Decoding Information 26

4.3 Why Does the Communication Method Work? 27

5 System Design and Implementation **29**
5.1 Development Environment 29
5.2 Communication-System Architecture 30
5.3 Encoder Implementation 31
5.4 Decoder Implementation 33
5.5 Implementation on SDR Hardware 36

6 Evaluation in the Simulation and Real-World Environments **42**
6.1 Parameters 44
6.2 Effect of the Number of Segments L 44
6.3 Communication Covertness 46
6.4 Effect of the Parameter γ 48
6.5 Results on SDR Hardware 50

7 Conclusion **54**

References **57**

List of Abbreviations

Abbreviation	English term
OFDM	Orthogonal Frequency-Division Multiple Access
MIMO	Multiple-Input Multiple-Output
BER	Bit Error Rate
GRC	GNU Radio Companion
SNR	Signal-to-Noise Ratio
MSNR	Mean Signal-to-Noise Ratio
PLS	Physical Layer Security
SDR	Software-Defined Radio
AWGN	Additive White Gaussian Noise
USRP	Universal Software Radio Peripheral
CRC	Cyclic Redundancy Check

Summary

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This thesis presents the design, implementation, and evaluation of a wireless communication system in which information is encoded in the statistical parameters of α-stable noise. An encoder and decoder were developed in GNU Radio Companion, initially evaluated through simulations, and subsequently deployed on USRP software-defined radio hardware. The effects of the asymmetry parameter β, scale parameter γ, number of segments L, and number of samples per symbol N on the bit error rate were analysed. The results show that an appropriate choice of N enables reliable transmission even under highly noisy conditions. Over-the-air measurements confirm the feasibility of the proposed approach while showing that reliability depends strongly on synchronisation and uninterrupted sample delivery.

Keywords: alpha-stable distribution, noise-based communication, software-defined radio, GNU Radio, physical-layer security.

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Chapter 1

Introduction

Wireless communication has become an indispensable part of everyday life. It enables mobile telephony, wireless Internet access, satellite navigation, and numerous industrial and Internet of Things (IoT) applications without which modern society is difficult to imagine. Most contemporary communication systems share the use of deterministic[1] signals, in which information is encoded by changing the amplitude, frequency, or phase of a carrier wave. Such approaches have proved highly effective, but they also have an important limitation. Because their signal structure is well known, suitable signal-processing methods can detect and analyse them and, in some cases, intercept them.

For this reason, **physical-layer security**[11] (PLS) is becoming increasingly important in modern networks. Rather than relying solely on cryptographic protocols at higher layers, the aim is to develop communication waveforms that are practically undetectable to an unauthorised eavesdropper. One promising approach is the use of stochastic[2] signals. In this approach, information is no longer encoded by changing the properties of a carrier electromagnetic wave, but in the statistical parameters of a random

[1] A deterministic signal is a signal whose value is fully known at every instant and can be described by an exact mathematical equation.

[2] A stochastic signal changes randomly over time and cannot be predicted exactly by a deterministic mathematical function; instead, it is described through statistical properties.

process. This represents a fundamentally different approach from that underlying conventional wireless communication systems.

The objective of this thesis is to develop a communication system based on α-stable noise in GNU Radio and deploy it on software-defined radio (SDR) hardware. The implementation is intended to make such a system usable on a standard SDR platform and thereby facilitate its use in further research and other wireless applications.

In addition to the implementation, the work experimentally evaluates the system through simulations and verifies its operation in an actual wireless transmission between two SDR devices.

Implementing such a system gives rise to several challenges:

- Estimating the parameters of α-stable distributions requires numerical methods because their probability density functions generally have no closed-form expression[3] (with only a few exceptions, including the Gaussian, Cauchy, and Lévy distributions). Parameter estimation is therefore more computationally demanding than methods based on classical distributions.

- Developing the communication system in GNU Radio requires new encoding and decoding blocks that allow α-stable noise to be used as the information carrier.

- Deploying the system on software-defined radio hardware requires adapting the implementation for real-time operation and addressing practical challenges such as time and frequency synchronisation, bandwidth limitations, and limited hardware computing resources.

This thesis addresses these challenges and presents an implementation of a communication system based on α-stable noise. Its main contributions are:

- Development of an encoder and decoder for communication using α-stable noise in GNU Radio Companion.

[3] An expression is in closed form if it can be written using only elementary functions.

- Implementation of the system on USRP software-defined radio hardware, enabling wireless data transmission in a real environment.
- Experimental analysis of how the system parameters affect the bit error rate (BER), including the effects of β, γ, and the number of samples per symbol.
- Demonstration of the communication system over an actual wireless link between two SDR devices.

To the best of our knowledge, this work presents the first open-source implementation of an α-stable noise-based communication system in GNU Radio Companion together with a demonstration on SDR hardware.

The results confirm that a communication system based on α-stable noise can be successfully implemented in GNU Radio Companion and deployed on software-defined radio hardware. The implementation provides a foundation for further research into communication systems that use stochastic signals as information carriers because it supports straightforward use and continued development in an open-source environment. This approach is particularly relevant where low probability of detection is important, for example in covert communication or critical-infrastructure protection. Its resistance to impulsive noise also makes it suitable for industrial environments in which conventional communication systems often struggle to provide reliable data transmission.

The source code for all implementations and the experimental results presented in this thesis are publicly available at `https://github.com/zanzan731/alpha-stable-noise-based-communication`.

Chapter 2

Wireless Communication and Its Challenges

Wireless communication systems transfer information without a physical transmission medium. They underpin mobile networks, wireless Internet access, satellite navigation, and many other technologies, although we rarely consider how information actually travels between devices. While wireless communication is now taken for granted, it is the result of decades of development in communication theory, modulation techniques, and radio technology that enable reliable data transmission through space.

This chapter introduces the basic principles of wireless communication, surveys the most widely used wireless technologies, and outlines the principal challenges faced by modern communication systems. Particular attention is given to the limitations of existing approaches that motivate the development of new communication systems.

2.1 Fundamentals of Wireless Communication

At first glance, transferring information without a physical connection between devices may appear almost self-evident because we use this form of communication every day. In reality, every wireless communication system

must exploit a natural phenomenon that allows information to propagate through space. Such transfer can be achieved in several ways, most commonly through magnetic fields or electromagnetic waves, including radio and optical transmission. Each method has its advantages and limitations. Magnetic fields support communication only over very short distances and are used mainly in technologies such as near-field communication (NFC). Optical transmission offers very high data rates but requires a direct line of sight between the transmitter and receiver. Radio transmission is therefore used most frequently in modern communication systems because it provides a favourable balance between range, data rate, and reliability. In the remainder of this thesis, wireless communication refers to the transfer of information by **electromagnetic waves** between a transmitter and a receiver without a physical transmission medium. An electromagnetic wave serves as the information carrier and propagates through the communication channel to the receiver. Because the wave itself contains no information, the information must be imposed on it through modulation. In conventional modulation, information is encoded in one or more properties of the electromagnetic wave: amplitude, phase, or frequency. Put simply, amplitude describes the height of the wave, phase its displacement, and frequency how closely successive waves follow one another. Once the signal has been encoded at the transmitter, it must be decoded at the receiver to recover the original message. This process is called demodulation. The modulation and demodulation process is illustrated in the wireless communication system diagram in Figure 2.1.

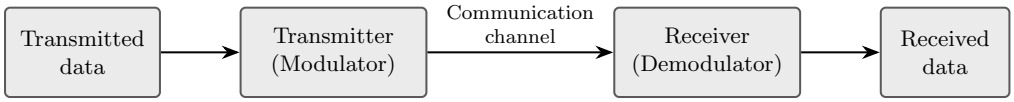


Figure 2.1: Diagram of a conventional wireless communication system.

Communication applications impose different requirements on range, data rate, power consumption, and interference resistance. Consequently, many wireless technologies and protocols have emerged, each optimised for a par-

ticular application. The following section presents the most widely used wireless technologies and their characteristics.

2.2 Existing Wireless Communication Technologies

Because applications have different requirements for range, data rate, power consumption, latency, and reliability, no single wireless system is suitable for every use case. Instead, several technologies have been developed and optimised for specific purposes. The most widespread include fifth-generation mobile networks (5G), Wi-Fi wireless local-area networks, Bluetooth, and LoRa systems. They differ primarily in range, data rate, energy consumption, and field of application. Table 2.1 summarises their main characteristics.

Technology	**Advantages**	**Limitations**	**Typical range**	**Common applications**
5G	High data rate, low latency, support for many users	Requires base-station infrastructure, higher power consumption	Up to several kilometres	Mobile networks, broadband access, industrial applications
Wi-Fi	High throughput, ease of use	Limited range, sensitivity to interference	Approximately 20–100 m	Home and enterprise local-area networks
Bluetooth	Low power consumption	Short range, lower data rate	Approximately 10–100 m	Wireless headphones, keyboards, smartwatches, sensors
LoRa	Very long range, very low power consumption	Low data rate	Up to several kilometres	Internet of Things (IoT), sensor networks, smart cities

Table 2.1: Comparison of commonly used wireless communication technologies.

As Table 2.1 shows, each technology is designed for particular requirements relating to data rate, range, power consumption, and reliability. Despite their differences, all listed technologies share one property: they transmit information by modulating deterministic signals, with information encoded in amplitude, frequency, phase, or a combination thereof. This approach underlies virtually all modern wireless communication systems.

Although these technologies have proved highly successful, they also have limitations. Their performance depends on channel conditions, especially noise, interference, and the possibility of detecting the communication itself.

The next section presents the key challenges of existing systems that motivate alternative approaches.

2.3 Key Challenges of Modern Wireless Communication Systems

Although modern wireless technologies have made exceptional progress in data rate, energy efficiency, and reliability, several challenges remain. They become particularly pronounced in demanding environments where noise is not adequately described by the usual additive white Gaussian noise (AWGN) assumption or where low detectability is as important as reliable transmission.

2.3.1 Noise

Noise is one of the most important factors affecting wireless communication quality. As a signal propagates through a channel, it is affected by natural and artificial sources of interference. The resulting distortion can cause the receiver to decode transmitted data incorrectly, increasing the bit error rate (BER).

Most contemporary communication systems are analysed and designed under an additive white Gaussian noise (AWGN) model. Owing to its mathematical simplicity, this model forms the basis of many modulation, detection, and coding methods and is suitable for numerous communication scenarios[18][9].

In many real environments, however, Gaussian noise is not an adequate model [12]. Industrial plants, power systems, and other electromagnetically demanding environments often contain impulsive noise characterised by rare but extremely strong disturbances. Such environments deviate substantially from the AWGN assumption, and systems based on that model may consequently suffer a marked degradation in performance.

2.3.2 Detectability of Communication

In addition to robustness, communication security is becoming increasingly important. Most modern systems employ cryptographic methods that effectively protect transmitted content, but encryption alone does not prevent the communication from being detected.

In applications such as military systems, critical-infrastructure protection, or certain industrial settings, it is desirable that an external observer cannot detect even the existence of the communication channel. If an attacker cannot detect the communication, jamming or targeted attacks on the system become substantially more difficult.

2.3.3 Existing Solutions

Various transmission techniques have been developed to reduce detectability and improve interference resistance, including frequency hopping and spread spectrum.

Frequency hopping is a technique in which the transmitter and receiver rapidly switch among frequency channels according to a predefined sequence. It reduces the effect of interference by allowing less noisy channels to be used and also makes interception more difficult[5].

Spread spectrum expands a signal over a wider frequency band than is strictly required to convey the information. Spreading the energy makes the signal more resistant to noise and interference and permits more reliable communication.

Although these techniques substantially improve resistance to interference and reduce the possibility of interception, they still rely on deterministic signals whose presence can be detected with sufficiently sensitive receivers and suitable signal-processing methods. Continued research into signals and communication methods that enable more reliable, secure, and less detectable information transfer therefore remains important.

2.4 Noise-Based Communication

The preceding sections treated noise as one of the principal factors degrading wireless communication. Most systems therefore seek to reduce its effect through suitable modulation, coding, and error-correction methods. This approach is used in virtually all modern wireless technologies.

Noise can, however, be viewed from a different perspective. Instead of treating it as an undesirable phenomenon, it can itself serve as the information carrier. This reverses the conventional view of wireless communication: noise is no longer merely a limitation but becomes a means of transmitting information. Information is then encoded not in the amplitude, frequency, or phase of a deterministic signal, but in the stochastic properties of a random process.

This approach opens new possibilities where conventional methods are not optimal. Because transmission is based on a random signal, communication is more difficult to detect, while channels dominated by impulsive noise can also be modelled more accurately.

Communication based on α-stable noise forms a distinct class of such approaches. The mathematical properties of α-stable distributions permit effective modelling of impulsive noise and the encoding of information in distribution parameters. Their fundamental properties are presented in the next chapter.

Chapter 3

α-Stable Distributions

Stable distributions form an important family of probability distributions that share several distinctive mathematical properties. Their foundations were systematically established by the French mathematician Paul Lévy in 1925, and they are therefore often called Lévy α-stable distributions.

Stable distributions, also known as α-stable distributions, provide a natural generalisation of the normal distribution. Unlike the normal distribution, they can model heavy tails[1] and asymmetry, enabling much more accurate models of phenomena with pronounced extreme values. These properties have led to important applications in telecommunications, signal processing, finance, and physics, where the classical Gaussian model often does not adequately describe the observed data.

3.1 Stable Distributions

The preceding chapters introduced stable distributions as the mathematical basis of the communication method considered in this thesis. Before presenting α-stable distributions in detail, it is useful to clarify what it means for a probability distribution to be stable.

[1] Heavy-tailed distributions assign extreme events a higher probability than the normal (Gaussian) distribution. Student's t-distribution is one well-known example.

Definition 1: Stable distribution

Let X_1 and X_2 be independent copies of a random variable X. The distribution of X is stable if, for arbitrary constants $a, b > 0$, there exist constants $c > 0$ and $d \in \mathbb{R}$ such that

$$aX_1 + bX_2 \stackrel{d}{=} cX + d.$$

Intuition

Intuitively, the definition states that the sum of two independent random variables with the same stable distribution again belongs to the same family. Addition changes only the location and scale parameters, while the shape of the distribution remains unchanged.

Stability is extremely important when modelling random processes. If a phenomenon consists of many independent contributions, it is desirable for their sum to remain in the same distribution family. The normal distribution is the best-known stable distribution, but it is not the only one. The following section therefore presents the family of stable, or α-stable, distributions and the parameters that determine their shape in greater detail.

3.2 Definition of α-Stable Distributions

Although all stable distributions share the stability property, their shapes differ. The family of α-stable distributions is fully specified by four parameters describing tail thickness, asymmetry, spread, and location.

All α-stable distributions considered in this thesis are absolutely continuous and possess a probability density function.

Definition 2: Absolutely continuous distribution

The distribution of a random variable X is absolutely continuous if there exists a non-negative integrable function p_X, called the *probability density function*, such that for every $x \in \mathbb{R}$,

$$F_X(x) = \mathbb{P}(X \leq x) = \int_{-\infty}^{x} p_X(t)\,\mathrm{d}t,$$

where F_X is the cumulative distribution function of X. Since the total probability of all possible values is equal to 1, it also follows that

$$\int_{-\infty}^{\infty} p_X(t)\,\mathrm{d}t = 1.$$

Unlike the normal distribution, the probability density functions of α-stable distributions generally cannot be written in closed form. They are therefore usually defined through their characteristic function.

Definition 3: Characteristic function

If a random variable X has probability density p_X, its characteristic function can be written as $\phi_X(t) : \mathbb{R} \to \mathbb{C}$:

$$\phi_X(t) = \mathbb{E}\left[e^{itX}\right] = \int_{-\infty}^{\infty} e^{itx}\, p_X(x)\,dx, \qquad t \in \mathbb{R}.$$

The characteristic function of an α-stable distribution is

Definition 4: Characteristic function of an α-stable distribution

$$\phi(t) = \begin{cases} \exp\left(i\mu t - \gamma^\alpha |t|^\alpha \left(1 - i\beta\ \mathrm{sign}(t) \tan\frac{\pi\alpha}{2}\right)\right), & \alpha \neq 1, \\ \exp\left(i\mu t - \gamma|t| \left(1 + i\beta\frac{2}{\pi}\,\mathrm{sign}(t) \ln\frac{\pi\alpha}{2}\right)\right), & \alpha = 1. \end{cases}$$

Because stability is not immediately apparent from the definition, we prove it below.

Proof 1: Stability of an α-stable distribution

Let X be an α-stable random variable, let X_1 and X_2 be independent copies of X, and let $a, b > 0$. Independence gives

$$\phi_{aX_1+bX_2}(t) = \phi_X(at)\phi_X(bt),$$

because

$$\phi_{aX_1+bX_2}(t) = \mathbb{E}\left[e^{it(aX_1+bX_2)}\right] = \mathbb{E}\left[e^{itaX_1}\right]\mathbb{E}\left[e^{itbX_2}\right] = \phi_{X_1}(at)\phi_{X_2}(bt).$$

For an α-stable distribution with $\alpha \neq 1$, the characteristic function is

$$\phi_X(t) = \exp\left(i\mu t - \gamma^\alpha |t|^\alpha \left[1 - i\beta\ \mathrm{sign}(t)\tan\left(\frac{\pi\alpha}{2}\right)\right]\right).$$

Substituting $t \mapsto at$ and $t \mapsto bt$ and using the independence of X_1 and X_2 gives

$$\begin{aligned}\phi_{aX_1+bX_2}(t) &= \phi_X(at)\phi_X(bt) \\ &= \exp\left(i\mu(a+b)t - \gamma^\alpha(a^\alpha + b^\alpha)|t|^\alpha \left[1 - i\beta\ \mathrm{sign}(t)\tan\left(\frac{\pi\alpha}{2}\right)\right]\right).\end{aligned}$$

We seek to write the resulting characteristic function in the same form as that of the random variable $cX + d$:

$$\phi_{cX+d}(t) = \exp\left(i(\mu c + d)t - \gamma^\alpha c^\alpha |t|^\alpha \left[1 - i\beta\ \mathrm{sign}(t)\tan\left(\frac{\pi\alpha}{2}\right)\right]\right).$$

Choosing $c = (a^\alpha + b^\alpha)^{1/\alpha}$ and $d = \mu(a + b - c)$ makes the resulting characteristic function exactly that of the random variable $cX + d$. Since a characteristic function uniquely determines a distribution, it follows that

$$aX_1 + bX_2 \overset{d}{=} cX + d.$$

This proves that the class of α-stable distributions is closed under linear combinations of independent identically distributed random variables, which

is precisely the property from which stable distributions derive their name.

3.3 Parameters of the α-Stable Distribution

The characteristic function in (3.2) shows that an α-stable distribution is specified by four parameters: the stability parameter α, asymmetry parameter β, scale parameter γ, and location parameter μ. Each controls a different property of the distribution.

3.3.1 Stability Parameter α

The stability parameter satisfies $0 < \alpha \leq 2$ and determines tail thickness. Smaller values of α produce heavier tails and thus a greater probability of extreme values. At $\alpha = 2$, the distribution becomes normal (Gaussian), with the thinnest tails in the α-stable family.

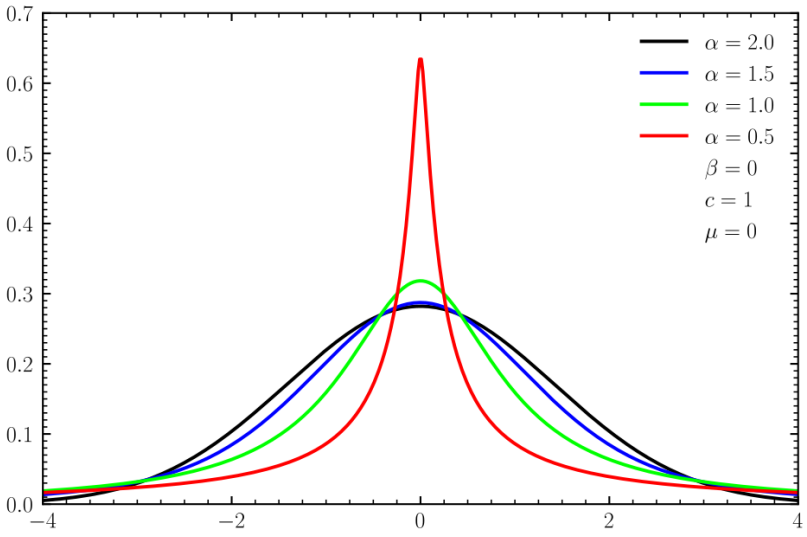


Figure 3.1: Distributions for different values of α [22].

3.3.2 Asymmetry Parameter β

The asymmetry parameter satisfies $-1 \leq \beta \leq 1$ and determines the magnitude and direction of asymmetry. The distribution is symmetric at $\beta = 0$; positive values produce a more pronounced right tail, whereas negative values produce a more pronounced left tail.

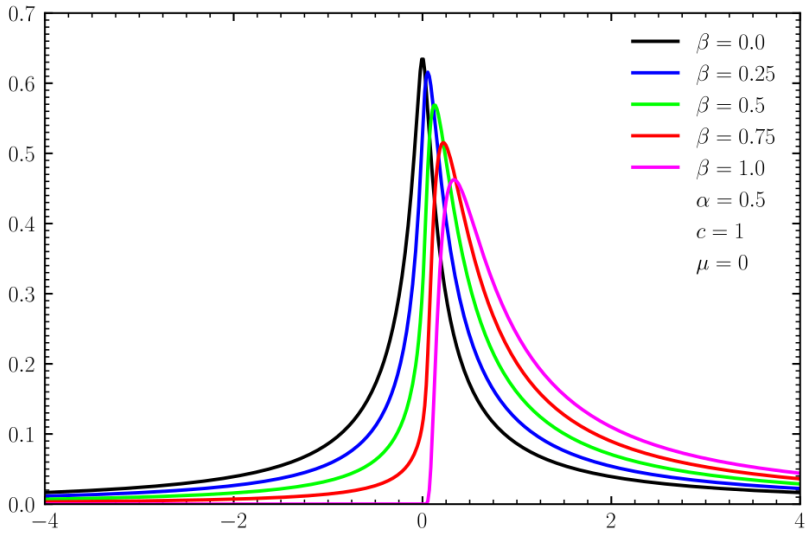


Figure 3.2: Distributions for different values of β [22].

The effect of negative values of β is obtained by reflecting the curves shown in the figure.

3.3.3 Scale Parameter γ

The scale parameter satisfies $\gamma > 0$ and determines the spread or scale of the distribution. Larger values produce a wider distribution. Its role resembles that of the standard deviation of a normal distribution, but the quantities are not identical because most α-stable distributions do not have finite variance.

3.3.4 Location Parameter μ

The location parameter may take any real value, $\mu \in \mathbb{R}$, and determines the distribution's position on the real axis. Changing μ shifts the entire distribution left or right without affecting its shape or spread.

3.4 Special Cases of α-Stable Distributions

Although α-stable distributions form a very general family, they include several well-known special cases that play important roles in mathematics, physics, and telecommunications. Some are listed in Table 3.1.

α	β	Distribution
2	0	Normal (Gaussian)
1	0	Cauchy
$\frac{1}{2}$	1	Lévy

Table 3.1: Special cases of α-stable distributions.

3.4.1 Normal (Gaussian) Distribution

At $\alpha = 2$, the result is the normal distribution, the only α-stable distribution with finite variance. Its density is symmetric and its tails are the thinnest within the α-stable family.

Its probability density can be written in closed form:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right),$$

where μ is the expected value and σ^2 is the variance.

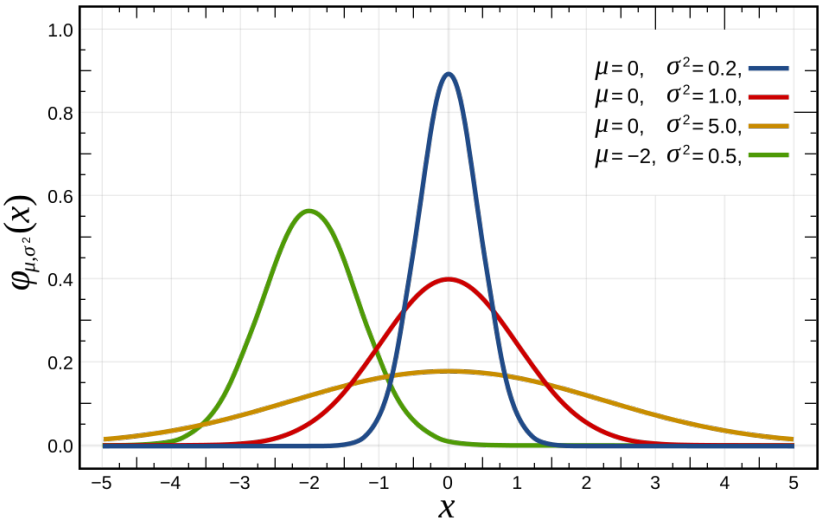


Figure 3.3: Probability densities of normal distributions for different parameter values [21].

3.4.2 Cauchy Distribution

The Cauchy distribution is the special case $\alpha = 1$ and $\beta = 0$. It is symmetric with very heavy tails and consequently has neither a defined expected value nor a finite variance. It is often used as an example of a distribution with a pronounced probability of extreme values.

Its density is

$$f(x) = \frac{1}{\pi} \frac{\gamma}{(x - x_0)^2 + \gamma^2},$$

where x_0 determines the centre of the distribution and γ determines its spread.

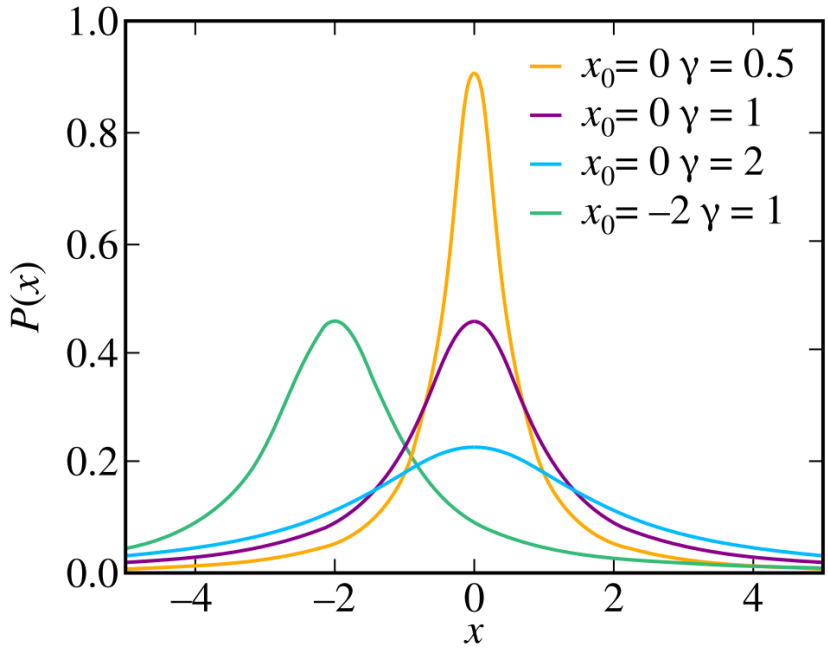


Figure 3.4: Probability densities of Cauchy distributions for different parameter values [19].

3.4.3 Lévy Distribution

The Lévy distribution is one of the most asymmetric α-stable distributions and corresponds to $\alpha = \frac{1}{2}$ and $\beta = 1$. Its very heavy right tail makes it suitable for processes in which rare but extremely large deviations may occur.

Its density is

$$f(x) = \sqrt{\frac{c}{2\pi}} \frac{\exp\left(-\frac{c}{2(x-\mu)}\right)}{(x-\mu)^{3/2}}, \qquad x > \mu.$$

Here, μ is the location parameter that shifts the distribution along the real axis, while c is the scale parameter that controls its spread.

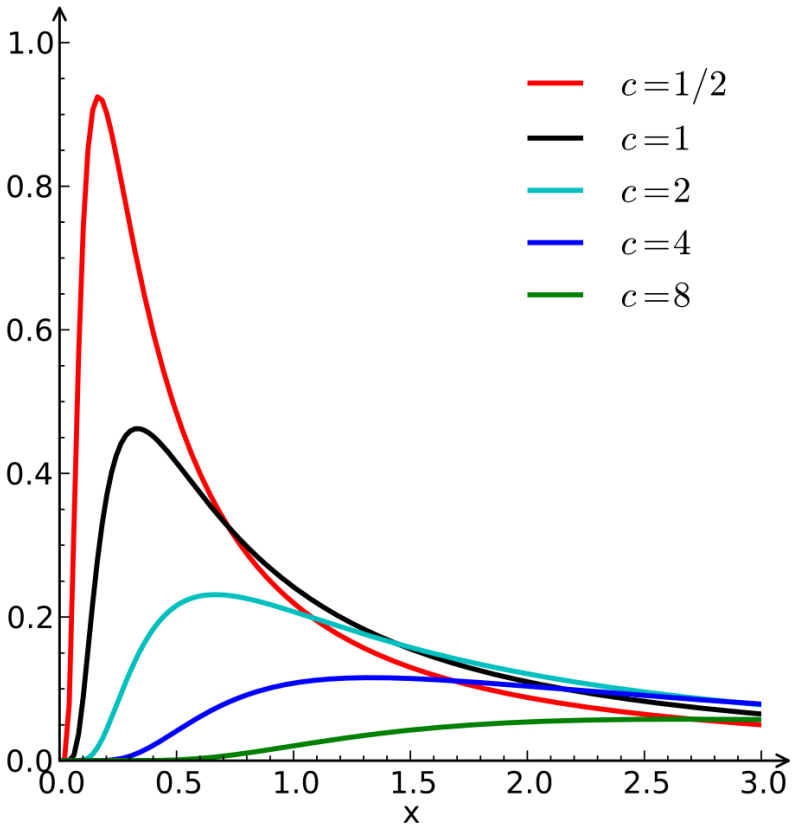


Figure 3.5: Probability densities of Lévy distributions for different parameter values [20].

3.5 Generalised Central Limit Theorem

The central limit theorem is one of the most important results in probability theory. It describes the behaviour of sums of many independent random variables.

Definition 5: Central limit theorem (CLT)

Let $X_1, X_2, \ldots, X_n$ be independent and identically distributed random variables with finite expected value

$$\mathbb{E}[X_i] = \mu$$

and finite variance

$$\mathrm{Var}(X_i) = \sigma^2 < \infty.$$

Then

$$\frac{X_1 + \cdots + X_n - n\mu}{\sigma\sqrt{n}} \xrightarrow{d} N(0,1),$$

where $\xrightarrow{d}$ denotes convergence in distribution.

Intuition

The cumulative distribution function of the normalised sum approaches that of the standard normal distribution. Importantly, the individual random variables need not themselves be normally distributed. This helps explain why the normal distribution occurs so frequently in nature and everyday phenomena.

The connection between the central limit theorem and α-stable distributions lies in their stability under the addition of independent random variables. The normal distribution is the special α-stable case

$$\alpha = 2.$$

For other parameter values,

$$0 < \alpha < 2$$

α-stable distributions have infinite variance, and for $\alpha \leq 1$ they do not

possess a finite expected value either. The assumptions of the classical central limit theorem, which requires a finite expected value and variance, are therefore not satisfied. The generalised central limit theorem shows that appropriately normalised sums of such random variables converge not to a normal distribution but to an α-stable distribution. In this sense, α-stable distributions provide a natural generalisation of the classical central limit theorem.

This connection is formalised by the generalised central limit theorem.

Definition 6: Generalised central limit theorem

Let Z be a random variable. Its distribution is stable if and only if there exist independent and identically distributed random variables

$$X_1, X_2, X_3, \dots$$

and constants

$$a_n > 0, \qquad b_n \in \mathbb{R},$$

such that

$$a_n(X_1 + \cdots + X_n) - b_n \xrightarrow{d} Z.$$

Intuition

If the sum of many independent and identically distributed random variables converges to a non-degenerate distribution after suitable scaling and shifting, that limiting distribution must be stable.

Finite variance is the key condition under which the classical central limit theorem yields a normal distribution. The generalised theorem extends this result to distributions with infinite variance. In that case, the limiting distribution is α-stable, and α determines the scaling rate of the sum:

$$a_n \propto n^{-1/\alpha}.$$

This is the generalised scaling factor. In the classical theorem, the factor is $\frac{1}{\sqrt{n}}$; substituting $\alpha = 2$ recovers exactly this expression.

Consequently, α-stable distributions are the only possible non-degenerate limiting distributions for suitably normalised sums of many independent and identically distributed random variables. The theorem explains why they are appropriate for signals formed from many random contributions: stability ensures that the statistical form of the signal is preserved when independent components are combined.

3.6 Significance of Stability for Communication-Channel Modelling

In a communication system, the signal received is generally not identical to the transmitted signal. During transmission it is affected by numerous independent random phenomena, including electromagnetic interference, multipath propagation, interference from other users, and noise from electronic components. The received signal is therefore often a sum of many random contributions.

It is therefore desirable for a mathematical channel model to remain in the same distribution family when independent components are added. This is explained by the stability property in Definition 1 and the generalised central limit theorem in Definition 6.

Mathematically, if the individual random contributions follow stable distributions, their sum again belongs to the same family. Only the parameters may change; the basic distributional form remains unchanged. This substantially simplifies system analysis because a single family can describe the complete channel regardless of the number of contributions forming the received signal.

Conventional systems usually assume that the individual contributions have finite variance. By the central limit theorem in Definition 5, the sum of many such contributions converges to a normal distribution, which explains the success of the AWGN model in many settings.

In many real environments, however, this assumption is not satisfied [12]. Impulsive noise contains rare but pronounced deviations and may not have finite variance. In such cases, the classical Gaussian model no longer adequately describes the channel.

This is where α-stable distributions become important. Stability keeps sums of many impulsive contributions within the same distribution family, enabling substantially more accurate models of channels dominated by impulsive noise. Beyond noise modelling, the property also suggests a fundamentally different communication method: rather than treating noise as undesirable, its statistical properties can be used to carry information. The communication method investigated in this thesis is based on this idea.

Chapter 4

Communication Using α-Stable Noise

The preceding chapter presented the mathematical properties of α-stable distributions. This chapter explains how those properties can be used to transmit information.

Conventional communication systems encode information in deterministic properties of a carrier, such as amplitude, phase, or frequency. In α-stable communication, information is represented not by the signal waveform but by the statistical parameters of its distribution.

Rather than specifying the instantaneous signal value, the transmitter generates a sequence of random samples from an α-stable distribution,

$$X_1, X_2, \ldots, X_n \sim S_\alpha(\beta, \gamma, \delta),$$

where the selected parameters represent the information to be transmitted.

4.1 Encoding Information

In communication using α-stable noise, information is represented not by an individual signal value but by parameters of the α-stable distribution from

which the signal is generated.

Definition 7: Theoretical operation of the transmitter

Let N be the number of samples representing one symbol. For each symbol, the transmitter generates the sequence

$$X_1, X_2, \ldots, X_N \sim S_\alpha(\beta, \gamma, \mu)$$

, where the selected distribution parameters determine the transmitted symbol.

An individual sample does not contain sufficient information about the underlying distribution because every sample is random. Only a sequence of samples reveals the distribution's statistical properties. Each symbol, or bit sequence, is therefore represented by N consecutive samples. Increasing N generally improves parameter estimation and reduces errors, but also increases the transmission time of each symbol.

Digital transmission requires a mapping between binary symbols and distribution parameters. Each input-bit combination is assigned a combination of α, β, γ, and μ. For each symbol, the transmitter selects the appropriate parameters and generates N samples, which the decoder subsequently receives and decodes.

4.2 Decoding Information

After transmission through the channel, the decoder receives N consecutive samples for each symbol. Its objective is to determine which α-stable distribution parameters were used to generate the sequence.

Definition 8: Theoretical operation of the receiver

Let N be the number of samples representing one symbol. For each symbol, the receiver obtains the sequence

$$Y_1, Y_2, \ldots, Y_N$$

, where

$$Y_i = X_i + W_i, \qquad i = 1, \ldots, N,$$

where X_i is the transmitted sample and W_i is channel noise.

Communication example

Assume that information is encoded only through β, with $\beta = -1$ representing bit 0 and $\beta = 1$ representing bit 1. For each bit, the transmitter generates N samples from the corresponding α-stable distribution. The receiver estimates β from the received samples. A positive estimate is decoded as bit 1; otherwise, the symbol is decoded as bit 0.

Decoding accuracy primarily depends on parameter-estimation quality. A larger N generally provides a better estimate and therefore more reliable recovery of the information.

4.3 Why Does the Communication Method Work?

Communication using α-stable noise relies on the mathematical properties introduced above. The key results are the stability property in Definition 1 and the generalised central limit theorem in Definition 6.

Stability ensures that a sum of independent α-stable random variables remains α-stable. The statistical structure is consequently preserved during generation and transmission, and the information remains encoded in the

distribution parameters rather than in any individual sample.

The generalised central limit theorem explains why distribution parameters can be estimated from a sample sequence. As N increases, the observed statistical properties become increasingly representative of the source distribution, and parameter estimates become more accurate.

If sufficiently distinct parameter combinations are assigned to different symbols, the estimated parameters can reliably distinguish them. A larger sample count therefore generally reduces the probability of an incorrect decision and improves transmission reliability.

Chapter 5

System Design and Implementation

This chapter presents the design and implementation of the developed communication system. It first outlines the development environment and system architecture, then describes the encoder and decoder, and finally presents the adaptations required to transfer the system from simulation to SDR hardware.

5.1 Development Environment

Software-defined radio (SDR) is a radio platform in which a large portion of signal processing is performed in software. The hardware primarily converts between analogue and digital signals and transmits or receives radio-frequency signals, while operations such as filtering, modulation, and demodulation run on a computer. The communication system can therefore be modified and tested without changing the radio hardware design.

In this thesis, SDR is used to transfer the developed system from simulation to an actual wireless link. The system could first be developed and evaluated in a controlled software environment and then verified through transmission between two radio devices. The fundamental encoding and decoding

logic remained unchanged, while the hardware implementation required additional adaptations for continuous sample processing and synchronisation.

The system was developed in the open-source GNU Radio environment [8] for processing data streams and digital signals. A system is assembled from interconnected functional blocks that exchange samples or other data. Each block performs a specific task, such as reading input, generating a signal, modelling a channel, or processing received samples. This block-based design allows individual components to be developed, tested, and replaced independently of the rest of the flowgraph.

GNU Radio Companion (GRC) was used to assemble and configure the blocks graphically. It represents the communication system as a flowgraph in which connections determine the direction of data transfer and block parameters determine processing behaviour. This view makes the complete system easier to inspect and permits rapid changes to simulation settings.

The encoder and decoder developed for this thesis are custom Python blocks because the required processing of α-stable distributions is not directly available in the standard GNU Radio blocks. The remaining components use existing blocks for reading and writing data, adding noise, and interfacing with radio hardware. The same environment could therefore be used both for simulation and for preparing the SDR implementation.

5.2 Communication-System Architecture

The developed system transmits information through the statistical properties of α-stable distributions. Its basic flow consists of a data source, encoder, communication channel, decoder, and data sink. Figure 5.1 shows the block diagram.



Figure 5.1: Block diagram of the communication system. The blocks shown in yellow were developed and implemented as part of this thesis, whereas the blocks shown in grey are existing GNU Radio blocks.

The data source produces a bit or data sequence and forwards it to the encoder. Based on the bit-to-symbol mapping, the encoder assigns appropriate α-stable parameters to every symbol and generates samples from the selected distribution.

The generated samples pass through the communication channel. In simulation, the channel is modelled in software and can include disturbances such as Gaussian noise; with SDR equipment, it is an actual radio link.

At the receiver, the decoder estimates the α-stable parameters from the received samples and uses them to determine the corresponding symbols or bits. The decoded data are passed to the sink, where they are reassembled into the output.

The following sections describe the encoder and decoder implementation in detail.

5.3 Encoder Implementation

The encoder converts an input bit sequence into a signal whose statistical properties represent the information. Unlike conventional systems, information is encoded not in the amplitude, phase, or frequency of a carrier, but in the parameters of an α-stable distribution.

The encoder receives a byte stream, converts it into individual bits, and stores them in a buffer. The number of bits required for one symbol is then

read from that buffer.

5.3.1 Mapping Bits to Parameters

The number of bits represented by a symbol depends on the number of available parameter values. For each parameter, the number of possible values must be a power of two so that bit combinations map uniquely to parameter values. The number of bits encoded by one parameter is

$n = \log_2(M)$

where M is the number of possible values. The total number of bits represented by a symbol is the sum of the numbers encoded by the individual parameters.

5.3.2 Mapping Symbols to Distribution Parameters

The implementation supports encoding information in α, β, and γ. In all simulations presented in this thesis, information is carried by β, while α and γ remain constant.

The input-bit combination determines an index in a mapping table from which the corresponding value of β is selected. Each symbol therefore represents a particular α-stable distribution.

5.3.3 Sample Generation

For each symbol, the encoder generates N samples from the α-stable distribution, where N corresponds to the variable `samples_per_symbol`. These samples form the encoder output and are passed to the communication channel or the next GRC block.

5.3.4 Chambers–Mallows–Stuck Algorithm

Procedure 1: Sample generation using the CMS algorithm

The signal is generated using the Chambers–Mallows–Stuck (CMS) algorithm [3][17],

$$X = \gamma C \frac{\sin(\alpha V + B)}{(\cos V)^{1/\alpha}} \left(\frac{\cos((1-\alpha)V - B)}{W} \right)^{\frac{1-\alpha}{\alpha}}, \tag{5.1}$$

where

$$B = \arctan\left(\beta \tan\frac{\pi\alpha}{2}\right), \tag{5.2}$$

$$C = \left(1 + \beta^2 \tan^2\frac{\pi\alpha}{2}\right)^{\frac{1}{2\alpha}}. \tag{5.3}$$

Here, V is uniformly distributed on

$$V \sim \mathcal{U}\left(-\frac{\pi}{2}, \frac{\pi}{2}\right), \tag{5.4}$$

while W is exponentially distributed with unit mean,

$$W \sim \mathrm{Exp}(1). \tag{5.5}$$

For every input symbol, the encoder thus produces N samples whose distribution is determined by the selected values of α, β, and γ.

5.4 Decoder Implementation

The decoder estimates the α-stable parameters from the received samples and reconstructs the original bit sequence. Because one sample is insufficient for reliable estimation, the decoder analyses several samples together.

5.4.1 Dividing a Symbol into Segments

The decoder stores the incoming sample stream in a buffer. Since the encoder generates exactly N samples for every symbol, individual symbols are

separated by processing each successive group of N samples.

5.4.2 Estimating the Parameter β

Procedure 2: Estimating the parameter β

The samples of one symbol are divided into L equal segments, each containing $K = N/L$ samples. For each segment, the maximum and minimum are found and their magnitudes are transformed logarithmically:

$$Y_{l,\max} = \ln\left(\max_{1\le k\le K} X_{l,k}\right), \qquad Y_{l,\min} = \ln\left(-\min_{1\le k\le K} X_{l,k}\right). \tag{5.6}$$

The means and variances are then calculated from the logarithmic extrema:

$$Y_{\max} = \frac{1}{L}\sum_{l=1}^{L} Y_{l,\max}, \tag{5.7}$$

$$S^2_{\max} = \frac{1}{L-1}\sum_{l=1}^{L}(Y_{l,\max} - Y_{\max})^2, \tag{5.8}$$

$$Y_{\min} = \frac{1}{L}\sum_{l=1}^{L} Y_{l,\min}, \tag{5.9}$$

$$S^2_{\min} = \frac{1}{L-1}\sum_{l=1}^{L}(Y_{l,\min} - Y_{\min})^2. \tag{5.10}$$

Here, $S_{\max}$ and $S_{\min}$ are the standard deviations, i.e., the positive square roots of the corresponding variances. They are first used to estimate α:

$$\hat{\alpha} = \frac{\pi}{2\sqrt{6}}\left(\frac{1}{S_{\max}} + \frac{1}{S_{\min}}\right). \tag{5.11}$$

The parameter β is then estimated using the hyperbolic tangent:

$$\hat{\beta} = \tanh\left(\frac{\hat{\alpha}}{2}(Y_{\max} - Y_{\min})\right). \tag{5.12}$$

The same estimator is also encountered in the equivalent exponential form

$$\hat{\beta} = 1 - \frac{2}{\exp(\hat{\alpha}(Y_{\max} - Y_{\min})) + 1}. \tag{5.13}$$

The following proof establishes the equivalence of the two forms.

Proof 2: Equivalence of the two forms of the β estimator

Let

$$z = \hat{\alpha}(Y_{\max} - Y_{\min}).$$

The definition of the hyperbolic tangent gives

$$\begin{aligned}\tanh\left(\frac{z}{2}\right) &= \frac{\exp(z/2) - \exp(-z/2)}{\exp(z/2) + \exp(-z/2)} \\ &= \frac{\exp(z) - 1}{\exp(z) + 1} \\ &= 1 - \frac{2}{\exp(z) + 1}.\end{aligned}$$

Substituting $z = \hat{\alpha}(Y_{\max} - Y_{\min})$ recovers the stated exponential form.

Thus, the hyperbolic-tangent and exponential forms of the β estimator are equivalent.

5.4.3 Bit Decoding

The estimate $\hat{\beta}$ is compared with the permitted values in the mapping table. The nearest value is selected, and its index represents the decoded bit combination. Decoded bits are buffered until eight are available, after which the decoder combines them into one byte and sends it to the output.

5.5 Implementation on SDR Hardware

Transferring the system from simulation to physical hardware introduces constraints that need not be considered in simulation. The most important are real-time sample delivery and synchronisation between transmitter and receiver.

The practical communication system used two USRP B210 devices [6] and two WA5VJB LP8565A2 log-periodic antennas [16]. The transmitter and receiver ran on a Lenovo Legion 5 Pro laptop and a desktop computer, respectively. The desktop computer was equipped with an Intel Core i7-10700 processor and 16 GB of RAM.

The USRP B210 is a dual-channel SDR device that supports simultaneous transmission and reception, covers the frequency range from 70 MHz to 6 GHz, and connects to the host computer through USB 3.0. The log-periodic antennas used in the experiment cover the frequency range from 850 MHz to 6.5 GHz and have a typical gain of 6 dBi, making them suitable for measurements at the selected frequency of 915 MHz.

In simulation, the sample-generation rate was not critical and the signal start was known exactly. A physical receiver, however, samples continuously, so the stream before a packet contains noise, interference, and possibly other environmental signals. The receiver must therefore determine the packet start autonomously.

An SDR transmitter must receive complex samples at the required sampling rate. If the computer cannot supply them quickly enough, an `underflow` occurs: the device temporarily runs out of new samples, potentially interrupting the signal and causing decoding errors. Since generating α-stable samples is more computationally demanding than generating simple deterministic signals, the generation method had to be adapted for real-time operation.

The following subsections present vectorised α-stable sample generation and the synchronisation procedure used between two SDR devices.

5.5.1 Vectorised Generation of α-Stable Samples

In the original simulation implementation, α-stable samples could be generated sequentially because execution time did not affect transmission continuity. This is unsuitable for SDR hardware, where samples must be supplied at a predetermined rate. The SDR implementation therefore generates samples in vectorised form. Before transmission begins, the complete packet is read and every input bit is mapped to a value of β.

In this implementation, information is encoded only through the asymmetry parameter β. Let M_β denote the number of possible values in its mapping table. One symbol then represents $\log_2(M_\beta)$ bits. Since every symbol contains N samples, transmitting B bits requires $\frac{BN}{\log_2(M_\beta)}$ samples. For the binary mapping $\beta \in \{-1, 1\}$, $M_\beta = 2$, so every symbol represents one bit and B bits require BN samples.

The parameter vector is obtained by repeating the value of β assigned to each symbol N times. Independent values $V_i \sim \mathcal{U}\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ and $W_i \sim \mathrm{Exp}(1)$ are then generated, with i running from 0 to $N-1$. Every sample therefore has independent V_i and W_i, while all samples belonging to one symbol share the same β. The samples are calculated using the CMS generation method in Procedure 1.

This does not make the samples of a symbol identical. Only β is shared; new random values V_i and W_i are used for every sample. Each sample is consequently an independent realisation of the selected α-stable distribution.

All samples are therefore produced in one vectorised operation. The complete transmission frame is then assembled and stored in the output buffer. GNU Radio can feed these precomputed samples gradually to the SDR without performing computationally expensive random-variable generation during transmission, substantially reducing the probability of underflow.

5.5.2 Transmission Packet Structure

Each transmission packet consists of a synchronisation sequence, a header repeated three times, a payload, and a guard tail. The synchronisation sequence contains 32 deterministic symbols and allows the receiver to detect the packet start.

The synchronisation sequence is followed by an 80-bit header containing a 16-bit identifier, a 32-bit payload-length field, and a 32-bit cyclic redundancy check (CRC). The header is transmitted three times, and the receiver determines its bits by majority voting to reduce the probability of decoding errors.

The header carries control information for packet processing rather than message payload. The identifier is a predefined constant known to both the transmitter and receiver. It allows the receiver to verify that a detected packet start belongs to the communication protocol in use. If the received identifier does not match the expected value, the receiver rejects the packet detection.

The length field specifies the number of bytes in the payload. In the measurements, the packet length was preset to 128 bytes, so the receiver could have obtained it from its own configuration. Nevertheless, including this field in the header permits packets of different lengths to be transmitted without configuring the receiver in advance.

After decoding, the CRC allows the receiver to check immediately whether the complete payload was received without detected bit errors. The CRC detects errors but does not correct them. The header therefore provides a technical basis for more reliable packet communication while supporting future protocol development and the addition of further control information.

The payload consists of pilot–data symbol pairs. Each α-stable data symbol is preceded by a deterministic pilot symbol, which the receiver uses to track carrier phase and frequency. Each pilot and data symbol contains N samples, where N is the configurable number of samples per symbol. For the binary mapping used here, transmitting B bits therefore requires $2BN$

payload samples.

Finally, a guard tail is transmitted and is not decoded. It prevents the end of the transmitted stream from affecting the last data symbols and also helps the receiver estimate the timing offset. Figure 5.2 shows the packet structure.

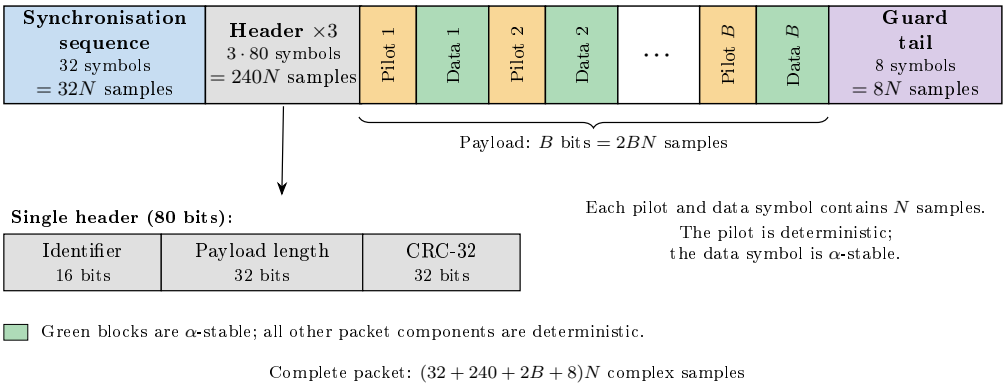


Figure 5.2: Structure of a transmission packet for communication over SDR hardware. The synchronisation sequence and header repeated three times are followed by pairs of pilot and α-stable data symbols and a guard tail.

5.5.3 Synchronisation

Synchronising an α-stable signal is difficult because it resembles noise. Searching for a packet start directly in the α-stable samples would therefore produce many false detections, as the receiver continuously observes noise and other interference even when the transmitter is idle.

A deterministic synchronisation sequence is therefore added to the beginning of every packet. The current sequence contains 32 bits, each represented by N samples. This permits more reliable correlation-based detection in noise. The transmitter generates the sequence in advance, while the receiver knows the same reference and searches for it in the incoming signal.

The receiver slides the reference sequence along the input stream and calculates signal correlation and coherence. A candidate is accepted only

when both exceed their configured thresholds, reducing the probability that random noise is interpreted as a packet start.

Thus, before every packet, the transmitter sends a predefined deterministic sequence that allows the receiver to detect the packet and determine its starting position.

5.5.4 Pilot Symbols and Carrier Tracking

Initial synchronisation determines the packet start and provides an initial carrier estimate, but this estimate is insufficiently accurate over a long packet. The transmitter and receiver use separate local oscillators, causing frequency and phase offsets, and the phase may also drift slowly during a packet.

The carrier phase cannot be estimated reliably from CMS-generated α-stable samples because they are too random. A deterministic pilot symbol is therefore placed before each α-stable data symbol.

For a given N, every bit is transmitted as N pilot samples followed by N α-stable samples. The pilot is always the same and carries no payload. Before subcarrier modulation, it consists of N samples with constant positive amplitude. The receiver estimates the local frequency offset from consecutive pilot samples and then the local phase offset from the mean phase of the frequency-corrected pilot. The resulting estimate corrects only the α-stable symbol immediately following that pilot.

This is more robust than using a single frequency and phase estimate for an entire packet because each data symbol receives a local carrier reference. The pilot does not resynchronise the packet or identify a new packet start; its sole purpose is local frequency- and phase-offset correction.

5.5.5 Clock-Offset Correction

Carrier frequency and phase offsets are not the only issues; the transmitter and receiver also have separate clocks. Although both devices are configured for the same nominal sampling rate, their actual rates differ slightly. This

error accumulates over a packet, so the final symbols of a long packet may shift by several hundred samples, making communication unusable.

A pilot alone does not remove the sampling-clock offset. The receiver therefore estimates the actual length of a pilot–data pair from the first transition between pilot and data and the transition between the last data symbol and the packet end. Based on this estimate, interpolation converts every pilot and data symbol to N samples. The method assumes that timing drift changes little during a packet or, if it changes, does so approximately linearly. The offset could be reduced by using a shared initial reference clock, but this would not be a realistic solution for the wireless system considered here because it would require a common timing reference between the transmitter and receiver. Furthermore, such equipment was not available for the experiment.

Chapter 6

Evaluation in the Simulation and Real-World Environments

This chapter evaluates the developed communication system in terms of decoding reliability, communication covertness, and resistance to noise. First, it analyses how the number of segments L affects the accuracy of estimating β and consequently the bit error rate. It then examines the trade-off between transmission reliability and communication covertness by varying the difference between the two values of β representing the individual bits. Next, it evaluates noise resistance for different values of γ and different numbers of samples per symbol N. Finally, it presents the SDR transmission results, thereby assessing the feasibility of the communication system in a real wireless environment.

All simulations used the system shown in Figure 5.1. The data source generated a random binary sequence, which the encoder converted into α-stable samples. The signal passed through a simulated channel, after which the receiver estimated β and reconstructed the transmitted bits. The stability parameter was fixed at $\alpha = 1.1$ in every experiment; the remaining settings are stated with the individual experiments.

Each configuration was evaluated in 100 independent repetitions, with 1024 randomly generated bits transmitted per repetition. Thus, every con-

figuration was evaluated over 102 400 bits. BER was calculated as the ratio of incorrectly decoded bits to all transmitted bits.

Because BER = 0 cannot be shown on a logarithmic axis, configurations with no detected errors are plotted at

$$\mathrm{BER} = \frac{1}{102\,400} \approx 10^{-5}.$$

This is the lower plotting limit and does not mean that one error was actually observed.

The simulation parameters are described before the results.

6.1 Parameters

Parameter	Description
`alpha_map`	List of possible values of α, currently reserved for future implementation.
`beta_map`	List of possible values of β. The current implementation encodes information exclusively through this parameter.
`gama_map`	List of possible values of γ. The current implementation does not use it to encode information, but it can be varied in experiments.
`samples_per_symbol`	Number of generated samples per symbol, denoted by N. A larger value generally improves estimation accuracy but increases the amount of transmitted data and simulation time.
`encode_{alpha,beta,gama}`	Selects the parameters α, β, and γ in which information is encoded. The current implementation uses only β.
`L`	Number of segments into which the decoder divides each symbol when estimating β. The estimate is calculated from the extrema of the individual segments.
`noise_ratio`	Amplitude of the added Gaussian channel noise. It determines the noise standard deviation in the GNU Radio *Noise Source* block; a larger value means more noise power and a lower signal-to-noise ratio (SNR)[7].
`source_number_of_samples`	Number of input bytes sent by the data source. It determines the message length and therefore the total number of symbols encoded and transmitted through the channel.

Table 6.1: Simulation parameters.

6.2 Effect of the Number of Segments L

To estimate β using Procedure 2, the decoder divides the N samples of a symbol into L equally sized segments, each containing $K = N/L$ samples.

This experiment compared $L \in \{2, 4, 5, 8, 10, 20\}$ for different values of N. The mapping was $\beta \in \{-1, 1\}$, with $\gamma = 1$ and `noise_ratio`$= 1$.

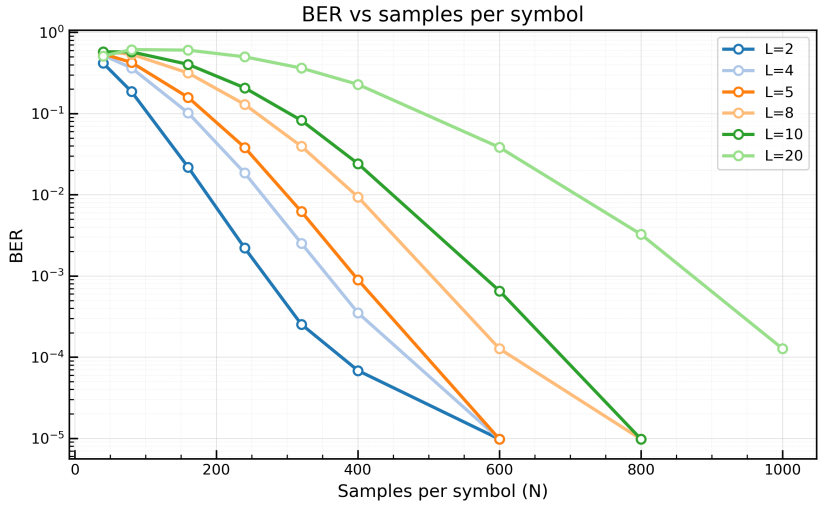


Figure 6.1: BER as a function of samples per symbol N for different values of L.

For every tested value of L, BER decreases as the number of samples per symbol increases. The decoder with $L = 2$ achieved the lowest BER throughout the observed range, whereas progressively larger values of L required more samples to attain comparable reliability.

The difference is particularly pronounced at $N = 400$. With $L = 2$, the BER was approximately $6.8 \cdot 10^{-5}$, corresponding to 7 errors among 102 400 transmitted bits. At $L = 8$, it was approximately $9.4 \cdot 10^{-3}$, or 963 errors—about 138 times as many as for $L = 2$. At $L = 20$, the BER reached approximately 0.23, corresponding to 23,414 errors, or about 3345 times as many as for $L = 2$. These absolute error counts make the difference, which is less obvious on a logarithmic plot, more tangible. No incorrectly decoded bits were observed for $L = 2$, $L = 4$, or $L = 5$ at $N = 600$, whereas for $L = 20$ the BER remained approximately $1.3 \cdot 10^{-4}$ even at $N = 1000$.

The simulation results show that, for the value $\alpha = 1.1$ used in this work, a smaller number of segments L generally produces a lower BER. This does

not, however, imply that a smaller L is always optimal. For a fixed number of samples N, increasing L provides more segments for statistical averaging but simultaneously reduces the number of samples $K = N/L$ in each segment. The optimal value of L therefore depends on α, the number of available samples, and the communication-channel conditions. For more impulsive distributions, particularly when $\alpha < 1$, a larger L may be preferable, whereas smaller values of L performed better at $\alpha = 1.1$ in the simulated environment considered here. It must also be noted that the simulated channel contained no additional `impulsive` interference. In an environment with rare and very strong disturbances, a larger L could reduce the effect of an individual interference impulse on the final estimate and thereby improve decoding. Consequently, L cannot be selected universally but must be chosen according to the statistical properties of the signal and the actual channel conditions [14].

6.3 Communication Covertness

The parameter β determines the asymmetry of an α-stable distribution and is used to encode information in the developed system. The experiment used symmetric mappings ranging from $\{-0.1, 0.1\}$ to $\{-1, 1\}$, with the difference defined as $\Delta\beta = \beta_1 - \beta_0$. A smaller $\Delta\beta$ means that the two bit distributions are more similar and therefore harder to distinguish. The difference may also be regarded as a security parameter because greater similarity makes decoding more difficult both for the intended receiver and for a potential eavesdropper.

The simulation used $L = 2$, $\gamma = 1$, and no additional Gaussian noise. The values $N \in \{40, 80, 160, 240, 320, 400, 600, 800, 1000\}$ were compared.

The results show that BER decreases as both $\Delta\beta$ and the number of samples per symbol increase. At small values of $\Delta\beta$, BER remains close to 0.5. In a binary system, this means that receiver decisions are not substantially more successful than random guessing. Communication with the

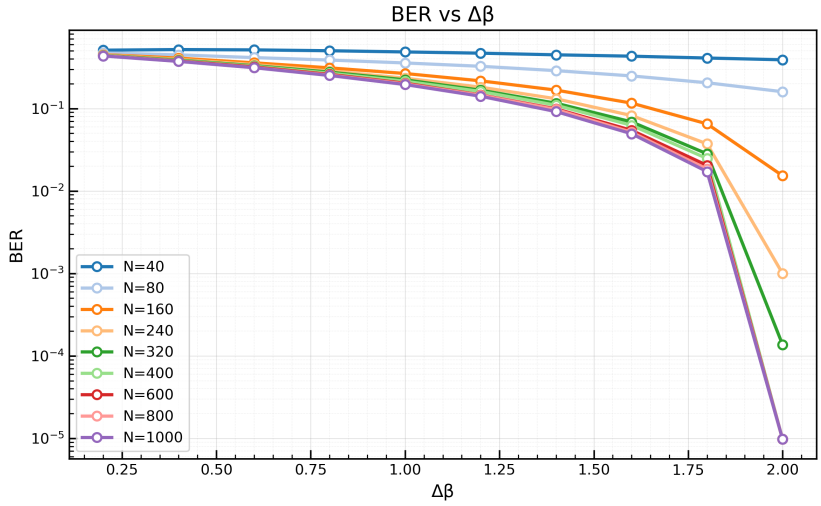


Figure 6.2: BER as a function of $\Delta\beta$ for different values of N.

current decoder is therefore practically unusable in this range.

The high BER is caused by the small difference in asymmetry between the distributions representing the two bits. Because only a finite number of samples is available and the estimate of β is subject to error, the two conditional distributions of the estimate $\hat{\beta}$ overlap substantially. The estimate therefore frequently falls on the wrong side of the decision threshold, resulting in incorrect decoding. As $\Delta\beta$ increases, the difference in asymmetry becomes more pronounced, the overlap between the estimates decreases, and the receiver can distinguish the symbols more easily.

The sharpest decrease in BER occurs at $\Delta\beta = 2$, corresponding to the mapping $\beta \in \{-1, 1\}$. At these extreme values, the two distributions are asymmetric in opposite directions, so their estimates of β are also better separated. Because the receiver estimates the parameter from the largest and smallest values in each segment, this mapping enables substantially more reliable decisions. At $N = 320$, BER decreased from approximately 0.028 at $\Delta\beta = 1.8$ to approximately $1.4 \cdot 10^{-4}$ at $\Delta\beta = 2$. For $N \geq 600$, no bit errors were observed at $\Delta\beta = 2$ in the completed repetitions.

These results characterise the performance of the current decoder and do

not necessarily represent a fundamental limit of the communication approach. A more accurate estimator of β that uses the information in the received samples more efficiently could reduce the dispersion of the estimates and enable more reliable communication even when $\Delta\beta < 2$. One possible direction for future work is to use a larger number of upper and lower extreme values or quantiles instead of only the maximum and minimum of each segment. Such a method would still need to be developed and evaluated experimentally. The effect of reducing $\Delta\beta$ cannot be eliminated completely because the two distributions become increasingly similar as the difference decreases. Nevertheless, an improved decoder could achieve a substantially better trade-off between BER and communication covertness.

6.4 Effect of the Parameter γ

This experiment analysed how γ and the number of samples per symbol N affect BER. The parameter γ controls the spread of the α-stable distribution and therefore the ratio between the information-signal power and the added Gaussian noise. The mapping was $\beta \in \{-1, 1\}$, with $L = 2$ and `noise_ratio`$= 1.0$. The values $N \in \{40, 80, 160, 240, 320, 400, 600, 800, 1000\}$ were compared.

MSNR was expressed in decibels as

$$\mathrm{MSNR}_{\mathrm{dB}} = 10 \log_{10}\left(\frac{\gamma}{\gamma_G}\right), \tag{6.1}$$

where $\gamma_G = 1$ is the scale parameter of the added Gaussian noise.

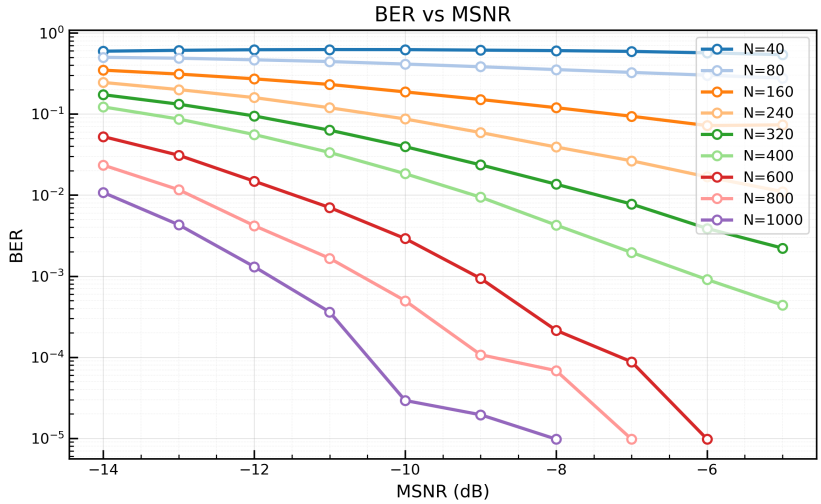


Figure 6.3: BER as a function of MSNR for different values of N.

The results show that BER decreases as both MSNR and the number of samples per symbol increase. The effect of N is particularly pronounced in noisy conditions. At MSNR $= -10\,\mathrm{dB}$, BER was approximately 0.62 for $N = 40$, $1.8 \cdot 10^{-2}$ for $N = 400$, $5.0 \cdot 10^{-4}$ for $N = 800$, and only $2.9 \cdot 10^{-5}$ for $N = 1000$. In the final case, only 3 of 102 400 transmitted bits were decoded incorrectly. Even at MSNR $= -14\,\mathrm{dB}$, $N = 1000$ achieved a BER of approximately $1.1 \cdot 10^{-2}$.

This behaviour indicates that the system can operate in considerably noisy environments. With a suitable channel model and adaptation rule, N could be selected according to the estimated noise level. When channel conditions deteriorate, the transmitter could increase the sample count to give the receiver more observations for estimating β; when conditions improve, N could be reduced again.

Such adaptation is relatively simple to implement because the mapping of β and the estimation procedure remain unchanged; primarily the number of generated and processed samples per symbol must change. Conventional adaptive systems often respond to changing conditions by switching among modulation and coding schemes, whereas the proposed system can retain its

basic encoding method and adjust only N.

Increasing the sample count also has a direct cost. At a fixed sampling rate, symbol duration grows in proportion to N, increasing transmission time and reducing the data rate. For the binary mapping used here, in which one symbol represents one bit,

$$T_s = \frac{N}{f_s}, \qquad R_b = \frac{f_s}{N}, \tag{6.2}$$

where f_s is the sampling rate, T_s is the symbol duration, and R_b is the bit rate. Selecting N therefore involves a trade-off between noise resistance and temporal efficiency, illustrated more intuitively below.

Example: effect of sample count on transmission rate

At a sampling rate of $f_s = 32\,000$ samples/s and $N = 1000$,

$$T_s = \frac{1000}{32\,000} = 31.25\,\text{ms}, \qquad R_b = \frac{32\,000}{1000} = 32\,\text{bit/s}.$$

If N is doubled to 2000, the symbol duration doubles to 62.5 ms and the bit rate halves to 16 bit/s.

6.5 Results on SDR Hardware

Measurements were performed using two USRP B210 devices and two WA5VJB LP8565A2 log-periodic antennas at a carrier frequency of 915 MHz and a sampling rate of 64 kS/s. The asymmetry parameter of the α-stable distribution was modulated using the bit mapping $\beta \in \{-1, 1\}$. The tested distances were 0.5 m, 1 m, and 2 m, with symbol lengths $N \in \{100, 200, 500\}$ samples. The value $L = 5$ was used in every experiment. For each combination of distance and N, one packet containing 128 bytes, or 1024 bits, was transmitted. No error-correcting code was used, so the calculated BER represents the raw bit error rate of the demodulator.

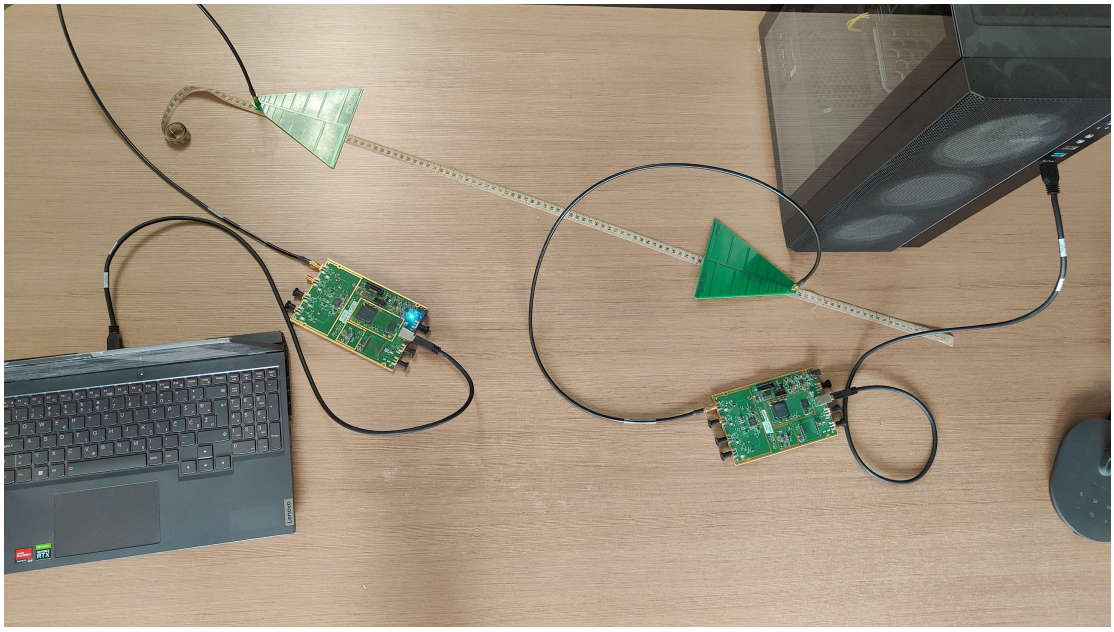


Figure 6.4: Experimental communication system comprising two USRP B210 devices, two WA5VJB LP8565A2 log-periodic antennas, and the transmitter and receiver computers.

BER was calculated by directly comparing the bits in the source and decoded files:

$$\mathrm{BER} = \frac{N_{\mathrm{err}}}{N_{\mathrm{all}}}. \tag{6.3}$$

Because each packet contained 1024 bits, one incorrect decision changes BER by approximately 0.0977 percentage points. The measurement results are summarised in the following table.

Distance	**Number of samples** N	**BER** [%]
0.5 m	100	4.688
	200	1.855
	500	0.195
1 m	100	3.125
	200	3.125
	500	0.293
2 m	100	4.297
	200	1.660
	500	1.172

Table 6.2: Measured BER as a function of distance and number of samples per symbol.

The measured values show that a larger number of samples per symbol generally improves decision reliability. With a larger N, the receiver has more realisations of the α-stable random variable available and can therefore estimate β with less statistical uncertainty. This is particularly evident at distances of 0.5 m and 1 m, where $N = 500$ achieved BER values of 0.195 % and 0.293 %, respectively.

The results are not fully monotonic with distance. For example, at $N = 200$, BER was lower at 0.5 m than at 1.0 m. Such deviations are expected because only one measurement was performed for each parameter combination. Each table entry therefore represents one realisation of a random communication process rather than an average over multiple repetitions. The result is affected by the particular realisation of the α-stable samples, the current environmental noise, multipath propagation, reflections in the room, and the instantaneous frequency and timing mismatch between the SDR devices. Because the packet is short, even a few additional incorrectly decoded bits can noticeably change the calculated BER.

Transmission reliability is also affected by synchronisation. After decoding, the receiver reports its estimate of the timing offset as the parameter `drift`. A large absolute value may indicate a difference between the sampling clocks, sample loss caused by underflow or buffer overflow, or an incorrect estimate of the packet end. If the estimated pilot–data pair length is incorrect, the timing-position error increases towards the end of the packet and a data symbol may also include part of a pilot symbol. The receiver implementation therefore rejects timing estimates that are inconsistent with the expected hardware accuracy and excludes samples at the edges of each data symbol during decoding.

The measurements confirm that data transmission by modulating β of an α-stable distribution is feasible on SDR hardware. They also demonstrate a trade-off between reliability and speed: a larger N improves the statistical estimate but lengthens the symbol and reduces the bit rate. A more reliable quantitative assessment of the effects of distance and N would require repeated measurements and reporting the mean BER with a confidence interval. Although underflow and overflow were not observed in the latest measurements, they remain possible on less powerful hardware or under more demanding conditions.

Chapter 7

Conclusion

This thesis developed and evaluated a wireless communication system in which information is transmitted by varying the asymmetry parameter β of α-stable noise. A custom encoder and decoder were implemented in GNU Radio Companion. Their operation was first evaluated in simulation and subsequently tested in an actual wireless transmission between two USRP B210 devices.

To the best of our knowledge, the main contribution of this work is the first open-source implementation of an α-stable noise-based communication system in GNU Radio that has also been tested in a real wireless environment. The implementation source code, system configurations, and measurement results are publicly available at `https://github.com/zanzan731/alpha-stable-noise-based-communication`, enabling experiment reproducibility and further development of the proposed approach.

The simulation results show that transmission reliability is strongly affected by the difference between the two values of β, the number of segments L, and the number of samples per symbol N. The lowest BER was obtained with the mapping $\beta \in \{-1, 1\}$ because the corresponding distributions differ most strongly and are easiest for the decoder to distinguish. However, this mapping is the least favourable in terms of communication covertness because the maximum separation between the distributions makes the signal

easier to distinguish and detect. Reducing the difference between the values of β increases distribution similarity and consequently communication covertness, but it also makes decoding more difficult and increases BER. Selecting β therefore involves a trade-off between transmission reliability and low detectability.

The SDR measurements confirmed that transmission is feasible without error-correcting coding. At $N = 500$, BER values of 0.195 %, 0.293 %, and 1.172 % were measured at distances of 0.5 m, 1 m, and 2 m, respectively. Increasing N was also found to improve reliability at the cost of a lower bit rate. Synchronisation and the uninterrupted delivery of samples between the computer and radio device emerged as the principal challenges in the practical implementation.

Substantial scope for improvement remains, particularly in the decoder. The current implementation uses a relatively simple method for estimating β together with basic synchronisation mechanisms, so the achieved results do not represent the upper performance limit of the communication approach. More robust estimation methods, adaptive decision thresholds, more accurate time and frequency synchronisation, and error-correcting codes could further reduce BER. An improved decoder could enable reliable operation at smaller differences between the values of β, increasing communication covertness, and at fewer samples per symbol, increasing the bit rate.

The developed implementation should therefore be viewed primarily as a foundation for further research rather than as the final limit of the system's usefulness. Future work could include developing a more advanced decoder, repeating measurements with more packets, improving synchronisation, and extending the encoding method to other parameters of the α-stable distribution. Such improvements could bring the system closer to practical use and substantially improve the trade-off among reliability, bit rate, and communication covertness.

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